Smart University Management Platform

An n8n-Based Workflow Automation System
AI-Powered Academic Support and Student Management

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Project: AI/LLM Summer School Capstone

Problem Statement

Universities manage large amounts of student and academic information, including assignments, deadlines, study schedules, and progress reports.

Managing these activities manually can lead to:

- Missed assignment deadlines
- Repetitive administrative work
- Difficulty organizing student records
- Limited visibility into academic progress
- Challenges in maintaining consistent communication

Need: A centralized workflow automation system to simplify routine academic tasks and improve information management.

Project Objectives

The main objective is to develop a workflow automation platform for routine university management tasks using n8n.
Specific Objectives

- Automate assignment reminder notifications.
- Organize student records using Google Sheets.
- Track pending assignments and their deadlines.
- Generate and deliver predefined study plans.
- Prepare and send weekly student progress reports.
- Reduce repetitive manual work through workflow automation.

Expected Outcome: A simple, modular prototype that connects data management and notifications through automated workflows.

Proposed Solution

The Smart University Management Platform uses n8n to connect academic data, workflow logic, and notifications.

The system includes five workflows:

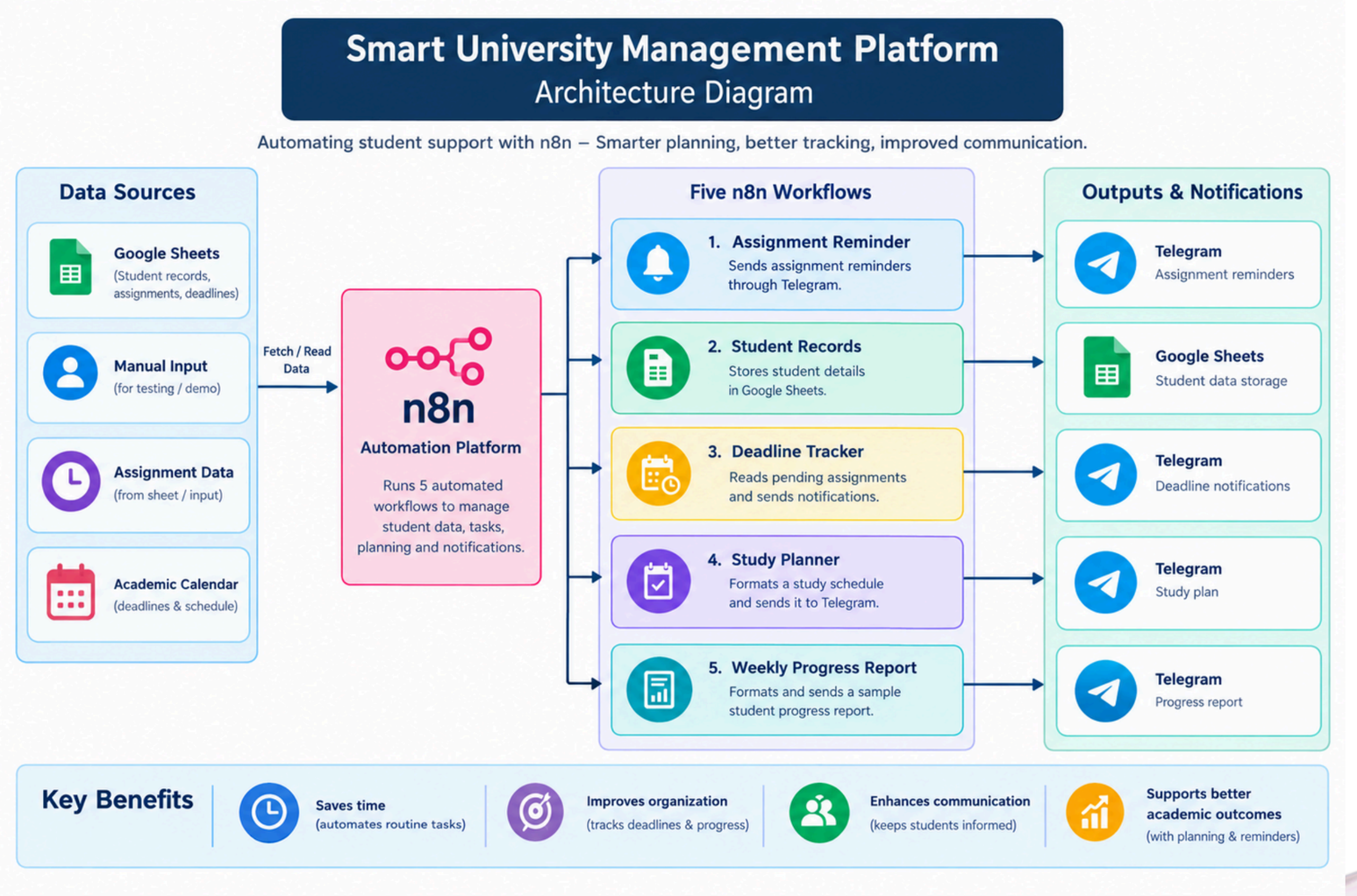
1. Assignment Reminder: Sends assignment reminders through Telegram.
2. Student Records: Stores student information in Google Sheets.
3. Deadline Tracker: Checks pending assignments and sends notifications.
4. Study Planner: Formats and delivers a predefined study schedule.
5. Weekly Progress Report: Formats and sends a sample student progress report.

How it works

Student and assignment data → n8n workflow automation → Google Sheets and Telegram outputs

The modular design allows each workflow to be tested and managed independently.

System Architecture



System Architecture

The platform uses n8n to coordinate five independent workflows.

Input Layer

- Sample student records
- Assignment details
- Study and progress information

Automation Layer — n8n

- Assignment Reminder
- Student Records
- Deadline Tracker
- Study Planner
- Weekly Progress Report

Output Layer

- Google Sheets — student and assignment records
- Telegram — reminders, study plans, and progress reports

Workflow 1: Assignment Reminder

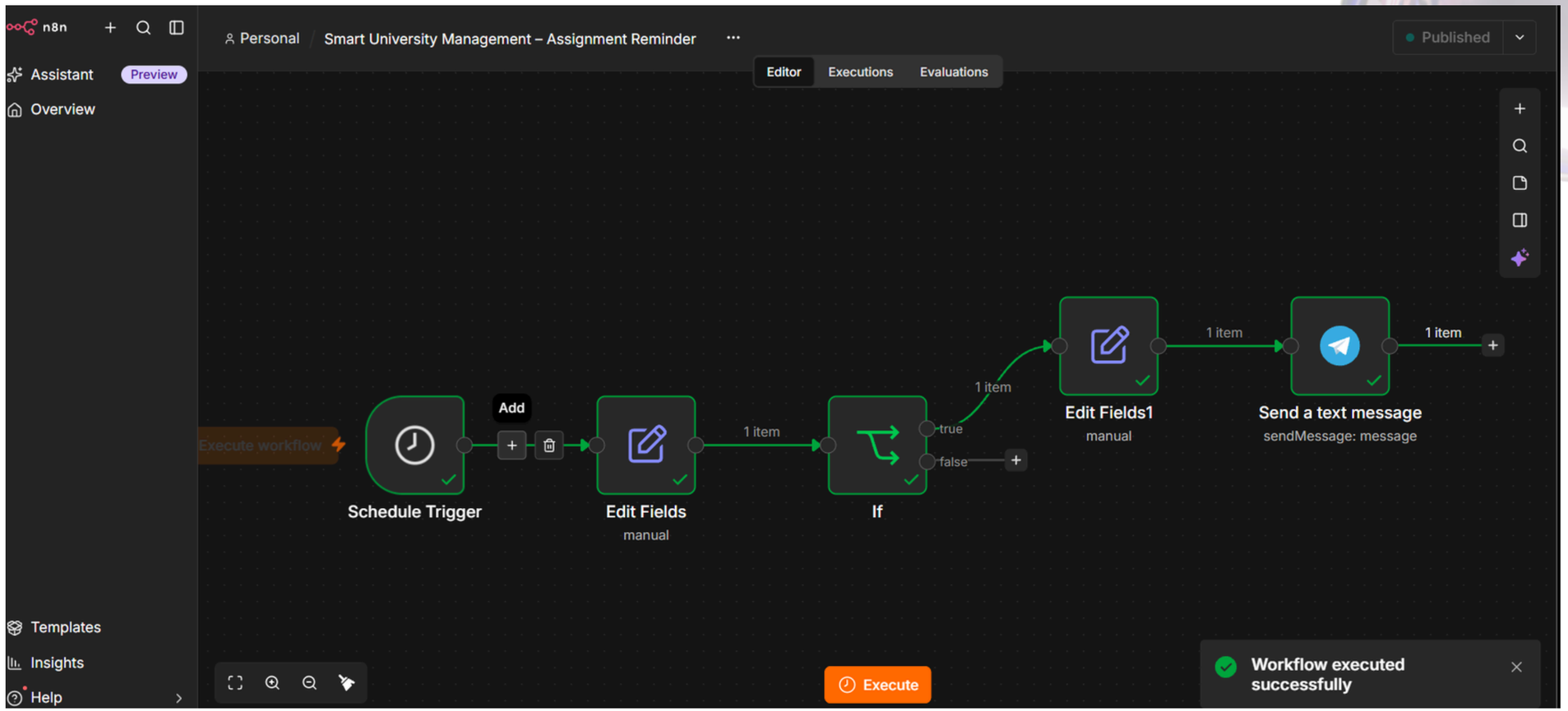
Purpose
Notify students about pending assignments through Telegram.

Workflow Steps

1. Start the workflow using a Schedule Trigger.
2. Enter assignment and student details.
3. Check whether the assignment status is Pending.
4. Generate a reminder message.
5. Send the reminder through Telegram.

Testing Result
Successfully tested the workflow and received the reminder message in Telegram.
Key Benefit: Helps students remember their pending assignments and deadlines.

Workflow 1: Assignment Reminder



Workflow 2: Student Records Management

Purpose: Store student information in an organized spreadsheet.

Workflow Steps

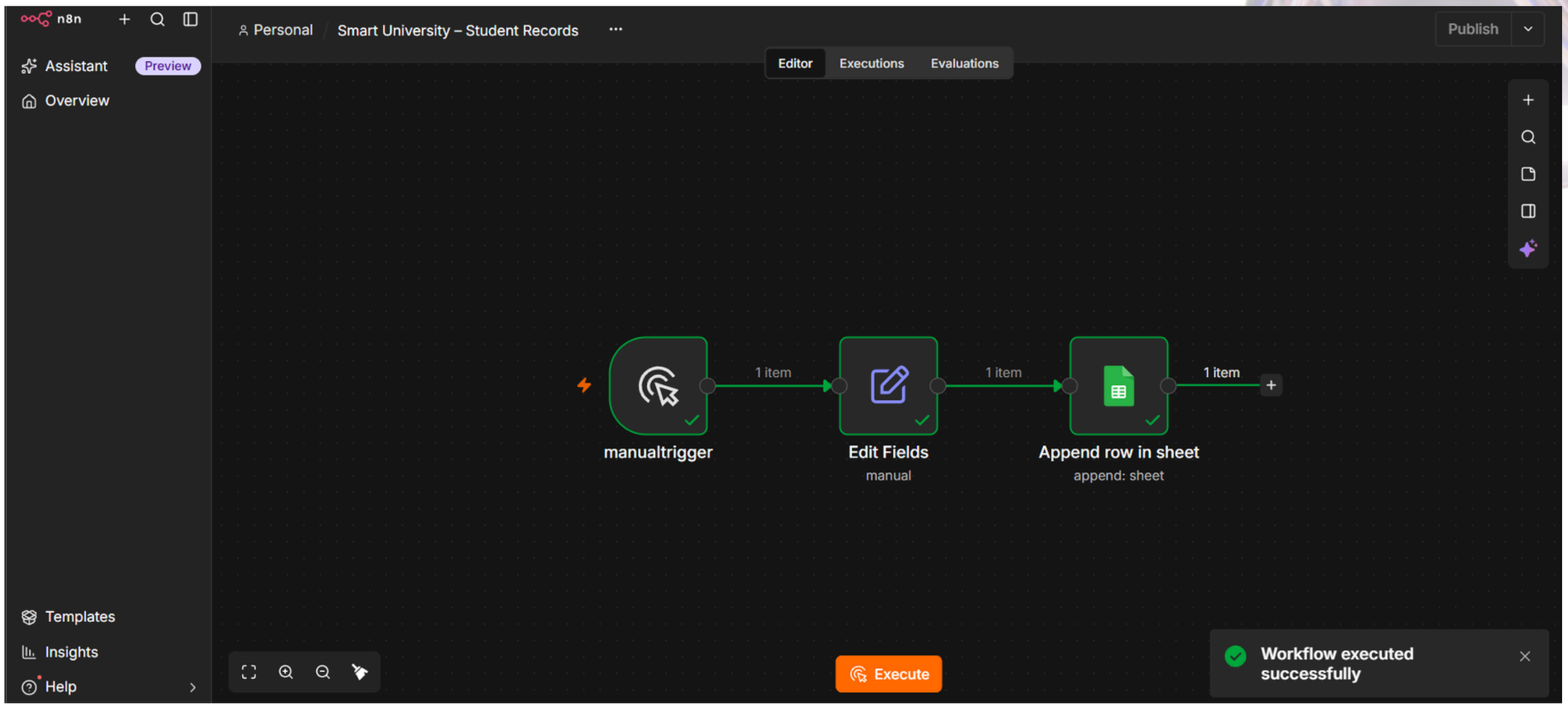
1. Start the workflow manually.
2. Prepare student details.
3. Append the data to Google Sheets.

Sample Fields

- Student Name
- Student ID
- Department
- Semester
- Email

Testing Result: Successfully added a sample student record to Google Sheets.
Benefit: Provides a simple, structured method for managing student information

Workflow 2: Student Records Management



Workflow 3: Deadline Tracker

Purpose: Identify pending assignments and notify students.

Workflow Steps

1. A scheduled trigger starts the workflow.
2. Assignment data is retrieved from Google Sheets.
3. An IF condition checks the assignment status.
4. Pending assignments trigger a Telegram notification.

Testing Result: Successfully retrieved assignment data and sent a Telegram notification for a pending assignment.
Current Scope: Checks assignment status; date-based deadline prioritization can be added in a future version.

Workflow 3: Deadline Tracker

Additional Academic Support Workflows

Workflow 4: Study Planner

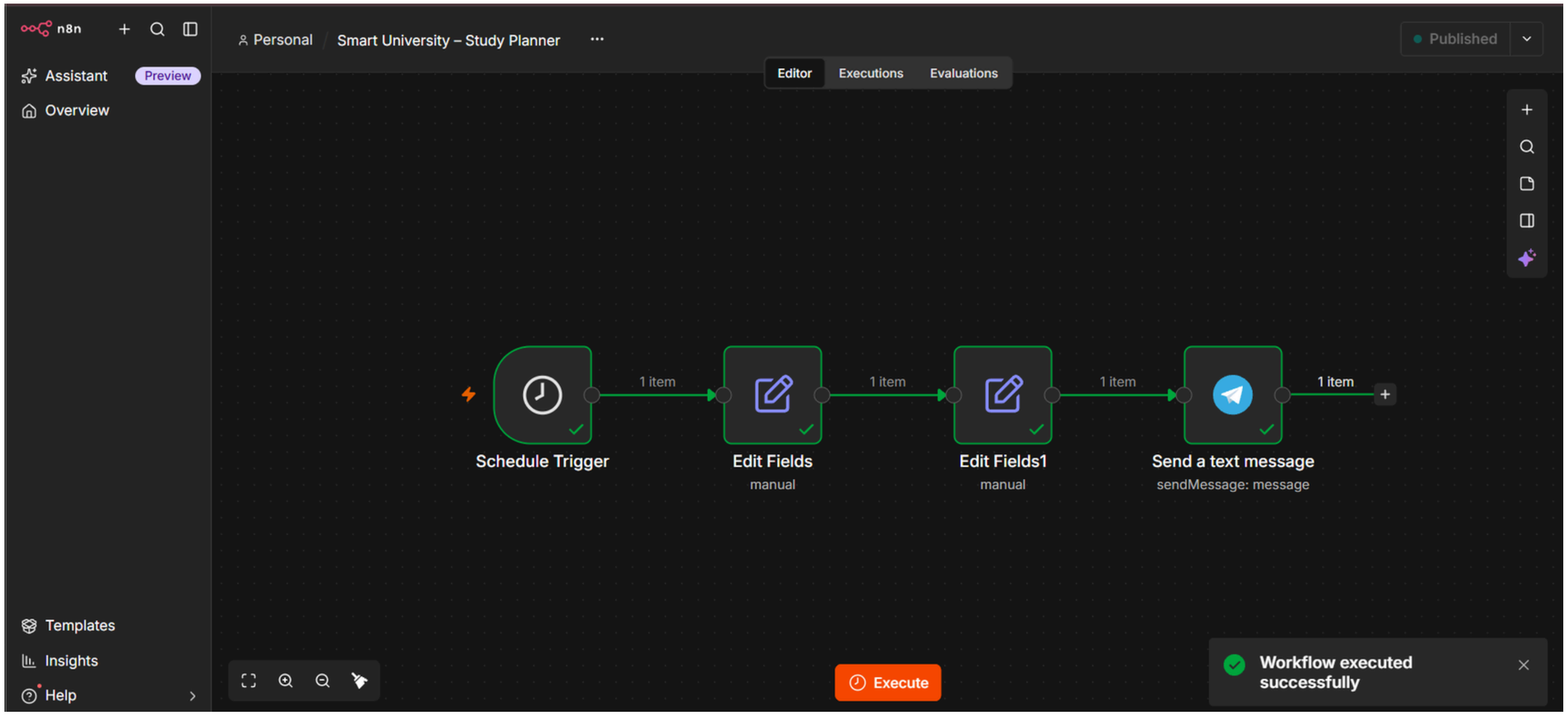
- Accepts student name, subjects, available hours, and priority.
- Formats a predefined study schedule.
- Sends the study plan through Telegram.
- Successfully tested with sample data.

Workflow 5: Weekly Progress Report

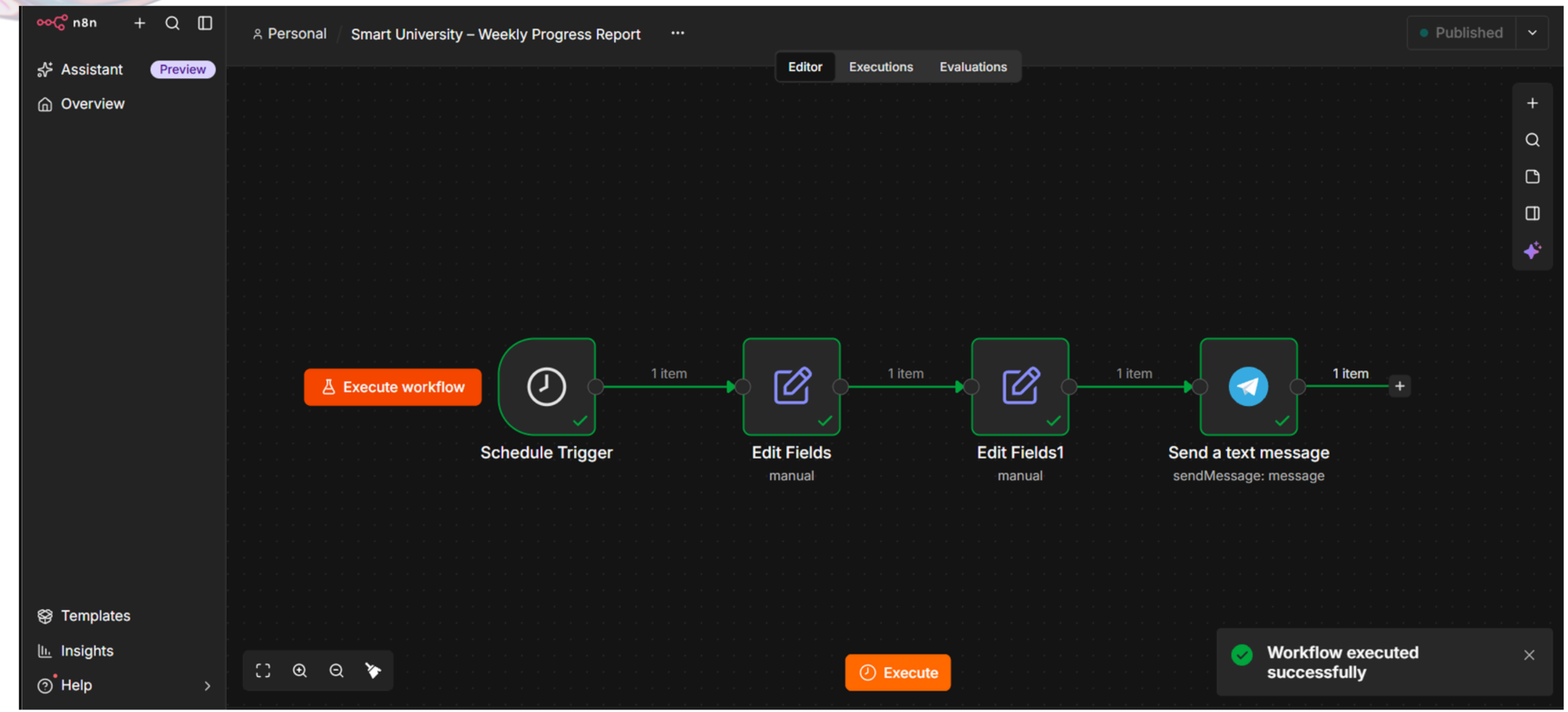
- Uses sample assignment and attendance information.
- Formats a weekly progress summary.
- Sends the report through Telegram.
- Successfully tested with sample data.

Benefit: Demonstrates automated delivery of personalized-format study plans and progress summaries.

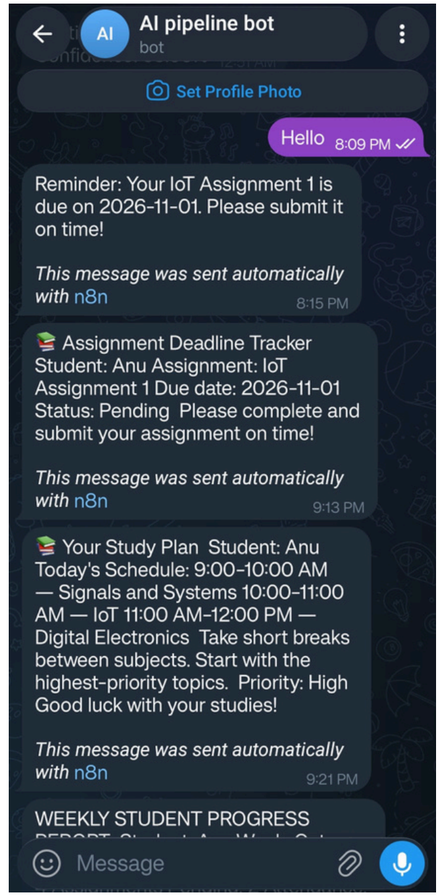
Additional Academic Support Workflows



Additional Academic Support Workflows



Results and Conclusion



Achievements

- Developed five n8n workflows.
- Connected Google Sheets for data management.
- Integrated Telegram for notifications.
- Tested workflow outputs using sample data.
- Exported workflows as JSON files for submission.

Current Limitations

- Uses sample student data.
- Study plans and progress reports use predefined values.
- Deadline tracking currently checks status rather than proximity to the due date.
- Student records are added through manual workflow execution.

Conclusion: The project demonstrates how workflow automation can simplify routine university management tasks.

Thank You